



Introduction to Dataproc

Hello, future data expert!

Welcome to this video all about Dataproc!

Together, we'll explore what makes it such a powerful tool and how you can implement it in order to both save time and increase the reliability of your data.

Let's begin with an example.

Maybe you work for a major online retailer that sells more than five hundred products and sources them from about 2,000 suppliers.

Obviously, this adds up to a lot of data —both structured and unstructured.

This retailer is working on a strategy to stay competitive in the fast-paced world of online retail.

Staying ahead of the competition is a top priority.

That's why your company is so focused on building a cutting-edge data system that revolutionizes the way they process data and customer reviews.

To help achieve this objective, you and your data team utilize Dataproc to manage the unstructured data.

Dataproc is a fully managed service that maximizes open-source data tools for batch processing, querying, streaming, and machine learning.

One of the benefits of Dataproc is that you can increase and decrease compute resources based on the project, so then you don't need to guess, underestimate, or overestimate resources.

This means your online retailer can dedicate cloud computing resources that focus only on processing customer reviews.

And you can have separate resources that focus only on processing sales information.

You also learn that Dataproc works especially well with Apache Hadoop, a framework that distributes the processing of data across clusters of computers.

This also means that we can scale up to use the power of multiple computers, even thousands, to process our data in real time when we need it.

Dataproc uses its open-source tools on virtual machines that scale up and down as needed.

And it can install a Hadoop cluster in 90 seconds.

Let's learn a bit about how Dataproc scaling works!

Dataproc disaggregates storage and compute services, so both can be created and terminated as needed.

This means that, once you have data to store, your storage will remain.

But if you're not actively processing data, the compute services will terminate.

For a more specific example, let's go back to our online retailer.

So, your company is going to store external application logs in cloud storage.

Then, the data will be processed by Dataproc, which can write it back to Google Cloud Storage or BigQuery.

You can also send the data to a data science notebook for analysis or to data scientists who can build and train AI models.

Because storage is separate from processing, your organization will save money by grouping tasks.

So, you'll be using exactly the right amount of memory and storage space to meet all your data analysis needs.

Your company has always stored data in different servers and on some hardware-based servers.

You learn that Dataproc has pre-built templates that can help you move your data from your existing structure into Google Cloud Services, or GCS.

Some of the common templates include Snowflake to GCS, Redshift to GCS, S3 To BigQuery, and Kafka to BigQuery.

This means your databases in Snowflake, Redshift, or Amazon can be automatically moved into Dataproc with these templates.

Plus, Dataproc makes it easier to manage Hadoop, and you can integrate organization-wide security.

Plus, you can enable data users through integrations, giving varied permissions to different users.

All right, now that your company has integrated Dataproc, it's able to provide data on all five hundred products to all of its vendors.

This has given your data team increased visibility into the best products to source and sell!

Best of all, your suppliers receive a smoother, consistent flow of valuable recommendations, enabling each vendor to tap into the waves of user demand like never before.

With real-time feedback and actionable insights, suppliers can fine-tune their product descriptions, ensuring they resonate with users on a whole new level!